QONTOS Scaled Quantum Architecture: A Scenario-Based Path to 1 Million Physical Qubits

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Abstract

This paper outlines the target QONTOS scaled quantum architecture under explicit base, aggressive, and stretch scenarios. The stretch-2030 scenario aims for **1,000,000 physical qubits** and **10,000 logical qubits** through a four-tier hierarchy of chiplets, modules, systems, and data centers. The paper does not claim that this architecture has been demonstrated today. Instead, it defines the architecture constants, subsystem assumptions, and validation gates that would need to be satisfied for the stretch scenario to become technically credible. In particular, the paper treats **100:1 effective physical-to-logical overhead**, millisecond-class superconducting qubits, high-rate photonic interconnects, and large-scale cryogenic deployment as linked program assumptions rather than settled results.

Claim status: architecture target and stretch scenario, informed by literature and QONTOS internal planning

Keywords: quantum computing, modular architecture, fault-tolerant computing, distributed quantum systems, scenario planning

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1. Introduction

QONTOS is pursuing a modular quantum computing architecture in which superconducting chiplets are integrated into cryogenic modules, connected into systems, and ultimately deployed as facility-scale quantum data centers. The architectural thesis is that modularity can relax the scaling bottlenecks of monolithic quantum processors, provided that qubit quality, error correction, interconnect performance, and cryogenic infrastructure all improve together.

The purpose of this paper is not to present a fully demonstrated million-qubit machine. Its purpose is to define the architecture and the conditions under which such a machine could become feasible.

1.1 Why Modular Scale Matters

The path to large-scale useful quantum computation is constrained by:

- chip yield and frequency crowding
- thermal load and wiring density
- error-correction overhead
- inter-module communication fidelity
- classical decoder and control latency

Monolithic architectures face increasing stress on all five fronts as qubit count rises. QONTOS therefore treats modularity as a first-class design principle rather than a late-stage packaging choice.

1.2 Scope and Claim Status

This paper uses the following claim labels:

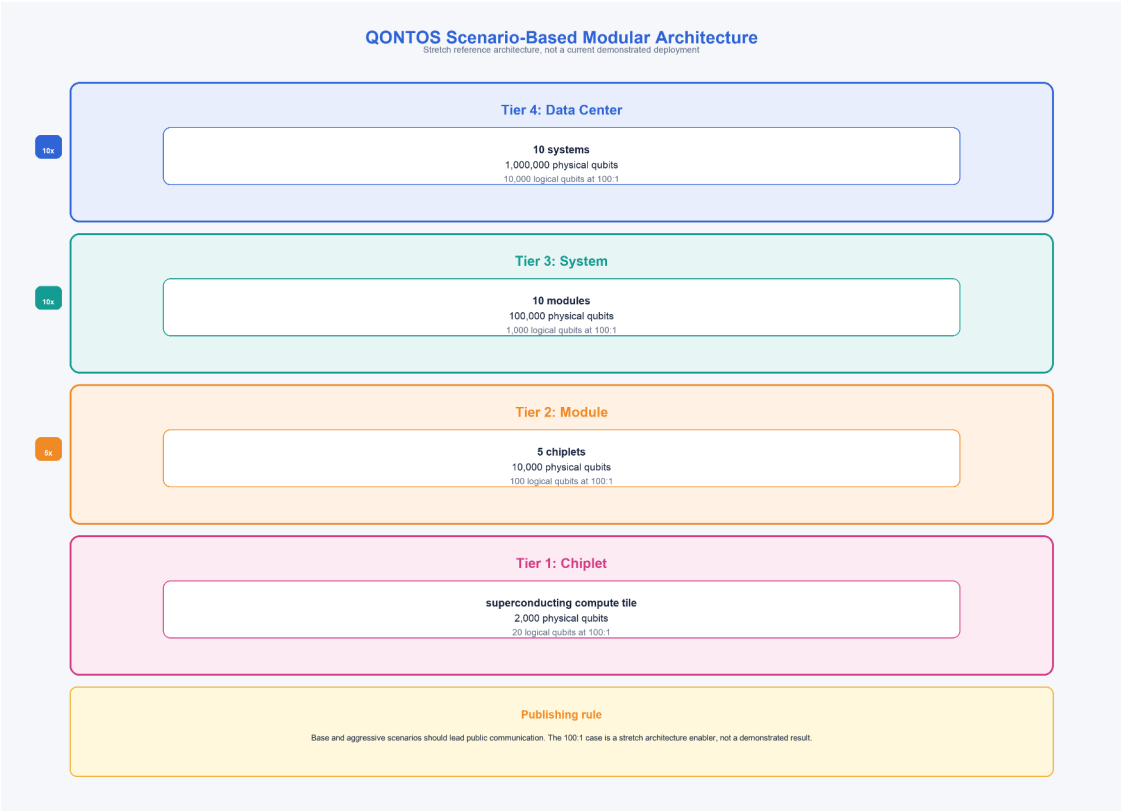
Label	Meaning
Demonstrated	Measured in QONTOS software or directly supported by published experimental results
Simulated	Supported by QONTOS digital-twin or simulator modeling
Derived from literature	Based on published external results or conservative extrapolation
QONTOS target	Internal engineering objective
Stretch target	Requires multiple major assumptions landing together

1.3 Architecture Claim Summary

Claim
QONTOS uses a four-tier modular hierarchy
2,000 physical qubits per chiplet
10,000 physical qubits per module
1,000,000 physical qubits by 2030
10,000 logical qubits by 2030
100:1 effective overhead

2. Canonical Architecture Constants

This section is the canonical source for architecture math across the research paper set.



Scaled Architecture

Figure 1: Target QONTOS four-tier hierarchy. The million-qubit configuration shown here is the stretch scenario, not a current demonstrated system.

2.1 Canonical Stretch Architecture

Layer
Chiplet
Module
System
Data center

2.2 Layer Definitions

Tier 1: QPU Chiplet

The chiplet is the fundamental fabricated superconducting qubit die.

Parameter	Stretch value	Notes
Physical qubits	2,000	Requires high-yield

		fabrication and manageable control density
Logical qubits at 100:1	20	Conditional on stretch QEC assumptions
Die size	15-20 mm	Placeholder target, not fabrication proof
Topology	Heavy-hex or equivalent sparse connectivity	Must remain compatible with control and error-correction assumptions

Tier 2: Quantum Module

The module is the cryogenic integration unit.

Parameter	Stretch value	Notes
Chiplets per module	5	Canonical constant
Physical qubits	10,000	Canonical constant
Logical qubits at 100:1	100	Stretch assumption
Inter-chiplet coupling	Microwave or equivalent short-range integration	Requires packaging and control validation

Tier 3: Quantum System

The system is the first photonic-connected machine layer.

Parameter	Stretch value	Notes
Modules per system	10	Canonical constant
Physical qubits	100,000	Canonical constant
Logical qubits at 100:1	1,000	Stretch assumption
Interconnect	Optical / photonic backbone	Requires transduction and entanglement validation

Tier 4: Quantum Data Center

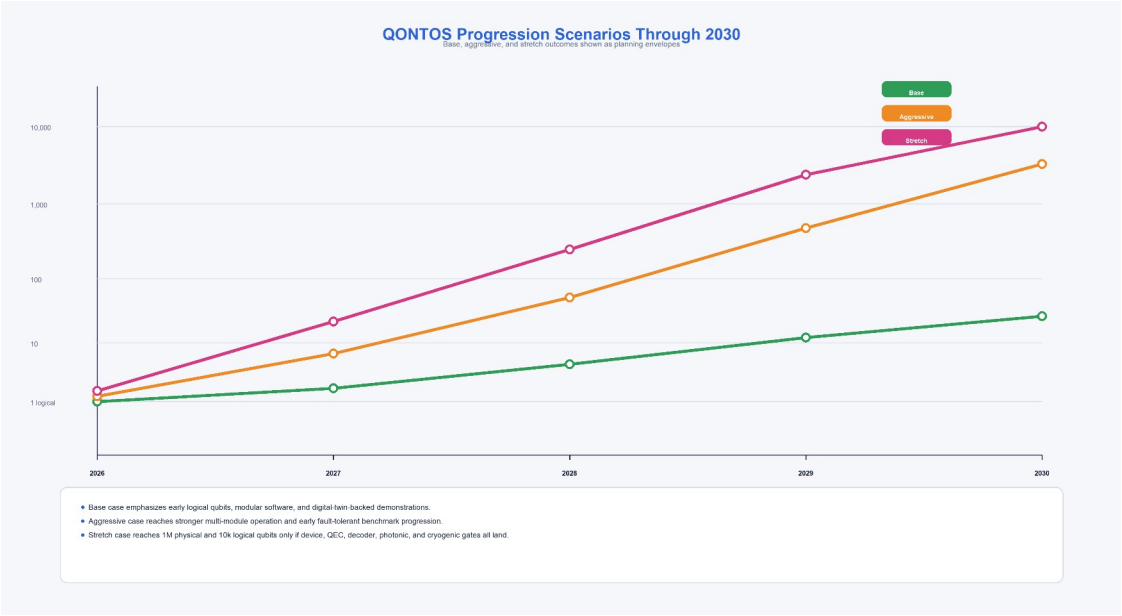
The data center is the facility-scale stretch endpoint.

Parameter
Systems per data center
Physical qubits
Logical qubits at 100:1
Power

Footprint

3. Scenario Framework

The million-qubit architecture should not be communicated as the only QONTOS future. This paper uses three scenarios.



Scenario Progression Timeline

Figure 2: Base, aggressive, and stretch scenario framing for QONTOS progression through 2030.

3.1 Base, Aggressive, and Stretch Cases

Metric
Physical qubits per chiplet
Physical qubits per module
Modules per system
Physical qubits per system
Systems per data center
Physical qubits per data center
Effective overhead
Logical qubits

3.2 Interpretation

- Base: early modular hardware plus strong software and digital-twin credibility

- Aggressive: large multi-module system with meaningful logical-qubit demonstrations
- Stretch: full million-qubit and 10,000-logical scenario

4. Error-Correction Envelope

4.1 Why 100:1 Is a Stretch Assumption

The transition from 1,000:1 to 100:1 effective overhead is central to the stretch architecture. However, this paper treats 100:1 as a contingent endpoint rather than a demonstrated property.

Overhead regime
1000:1
300:1
100:1

4.2 System Implications

For 10,000 logical qubits:

- At 1000:1 overhead: 10,000,000 physical qubits
- At 300:1 overhead: 3,000,000 physical qubits
- At 100:1 overhead: 1,000,000 physical qubits

This is why the million-qubit architecture and the 100:1 claim must always be presented together as a coupled stretch scenario.

4.3 Enabling Technologies

The stretch architecture depends on all of the following improving together:

1. millisecond-class qubits
2. high two-qubit fidelity at scale
3. low-overhead fault tolerance
4. fast decoder and control loops
5. photonic interconnect performance sufficient for modular operation

5. Roadmap Scenarios

5.1 Five Program Phases

Phase	Years	Base outcome	Aggressive outcome	Stretch outcome
FOUNDATION	2025-2026	improved software, digital	first small hardware	first evidence toward

		twin, early device validation	validation	millisecond-class devices
SPUTNIK	2026-2027	small modular hardware	scaled cryogenic module	10,000 physical qubits and early logical milestones
PIONEER	2027-2028	multi-module simulation and software maturity	hardware-linked distributed runtime	100,000 physical qubits and 1,000 logical qubits
HORIZON	2028-2029	robust modular control and benchmarks	meaningful FTQC demos	500,000 physical qubits and 5,000 logical qubits
SUMMIT	2029-2030	commercially useful sub-stretch system	large-scale logical platform	1,000,000 physical qubits and 10,000 logical qubits

5.2 Validation Gates

Gate	Why it matters
G1: device metrics cross agreed threshold	architecture cannot scale without qubit quality
G2: first logical qubits on modular substrate	roadmap cannot rely only on raw physical qubits
G3: inter-module entanglement and scheduling work in practice	architecture fails if modular links underperform
G4: digital twin and benchmark harness match experimental trajectory	roadmap becomes guesswork without this
G5: economics remain plausible under actual overhead and facility assumptions	million-qubit narrative fails if system cost explodes

5.3 No-Go Conditions

The stretch scenario should be downgraded if any of these occur:

- effective overhead remains closer to 1000:1 than 300:1
- photonic transduction stays in low-single-digit efficiency
- large chiplet packaging proves unmanageable
- cryogenic and control budgets fail to scale with module size

6. Performance Envelope

6.1 Scenario-Based Performance Table

Metric
Physical qubits
Logical qubits
Effective overhead
2Q fidelity
Transduction efficiency
Decoder latency

6.2 Competitive Framing

This paper should not present precise competitor 2030 numbers as settled facts. Instead:

- describe monolithic versus modular architectural differences
- compare scenario classes rather than exact future qubit totals
- reserve exact competitive comparisons for the benchmarking methodology paper

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7. Strategic Implications

If the stretch architecture becomes feasible, it would move QONTOS from an orchestration-layer company with a modular thesis to a full-stack modular quantum systems company. If the stretch architecture does not fully materialize, the aggressive and base scenarios still support a credible business in modular software, digital twin tooling, benchmarking, and early hardware integration.

This is why the scenario framing is important: it protects the core thesis even if the stretch case arrives later than hoped.

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8. Conclusion

QONTOS has a coherent modular scaling thesis, but the million-qubit architecture should be understood as a stretch scenario built on multiple linked assumptions. The value of this paper is therefore not that it proves million-qubit delivery today. Its value is that it defines a technically structured target architecture, a canonical system ladder, and a set of validation gates against which progress can be measured.

The most credible near-term message is:

- QONTOS has a software-first and modular-native foundation today
- QONTOS has an aggressive research roadmap toward large-scale modular quantum systems

- the million-qubit / 10,000-logical path remains a stretch target contingent on subsystem breakthroughs

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Claim posture: Architecture target with base/aggressive/stretch scenarios